Natural Computing

LMU Munich
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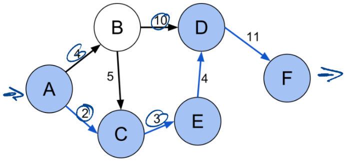
Definition 9 (optimization). Let $\mathcal{X}$ be an arbitrary state space. Let $\mathcal{T}$ be an arbitrary set called target space and let $\leq$ be a total order on $\mathcal{T}$. A total function $\tau : \mathcal{X} \to \mathcal{T}$ is called target function. Optimization (minimization/maximization) is the procedure of searching for an $x \in \mathcal{X}$ so that $\tau(x)$ is optimal (minimal/maximal). Unless stated otherwise, we assume minimization. An optimization run of length $g + 1$ is a sequence of states $\langle x_t \rangle_{0 \leq t \leq g}$ with $x_t \in \mathcal{X}$ for all t.

often: $\mathcal{T} = \mathbb{R}$

$\rightarrow \langle x_0, \dots, x_g \rangle$

Shortest Path



$$\mathcal{X} = \mathcal{P}(E) = 2^E$$

for graph edges E

$$\mathcal{T} = \mathbb{Z}$$

$$\tau = \{e_0, \dots, e_n\} \longmapsto \sum_{i=0}^{n} cost(e_i)$$

$$\tau : \mathcal{X} \to \mathcal{T}$$

example

$$x = \{(A,C), (C,E), (E,D), (D,F)\}$$

$$\boxed{x^* = \arg\min_{x \in \mathcal{X}} \tau(x)}$$

source: en.wikipedia.org/wiki/Shortest_path_problem#/media/File:Shortest_path_with_direct_weights.svg

Linear Regression

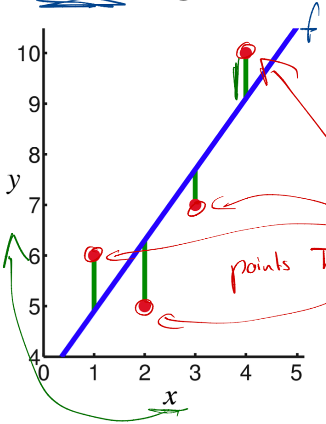
$\mathcal{X} = \mathbb{R}^2$

so that $f(x) = ax + b$ for $x = (a,b)$

$x \in \mathcal{X}$

$\mathcal{Y} = \mathbb{R}$

$\tau((a,b)) = \sum_{i=0}^{|T|-1} |(ax_i + b) - y_i|$



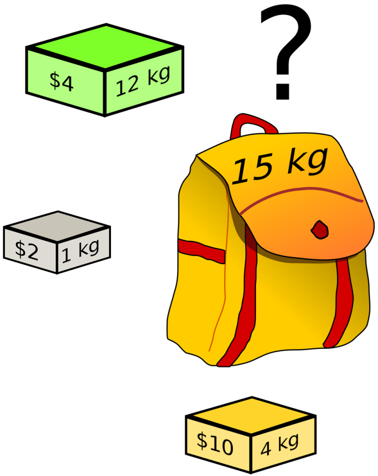
points $T = \{(x_0,y_0), \ldots, (x_3,y_3)\}$

$\mathcal{X} = \{ f: \mathbb{R} \to \mathbb{R} \mid f \text{ is linear} \}$

↳ mathematically possible but very inefficient

source:
en.wikipedia.org/wiki/Linear_regression#/media/
File:Linear_least_squares_example2.png

The Knapsack Problem (NP-complete)



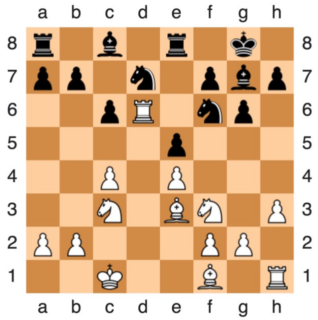
$$\tau : \underbrace{\mathcal{X}}_{\mathcal{X} = \wp(\text{Items})} \longrightarrow \underbrace{\mathcal{T}}_{\mathcal{T} = \mathbb{N}}$$

Items = { (\$4, 12kg),
(\$2, 2kg),
⋮
(\$10, 4kg) }

maximize $\tau(\{(d_1, w_1), \dots (d_n, w_n)\}) = $

$$= \begin{cases} 0 & \text{if } \sum_{i=1}^{n} w_i > 15 \\ \sum_{i=1}^{n} d_i & \end{cases}$$

Playing Chess



$\mathcal{X} = \mathbb{R}^{100\,000}$ weights in a neural network

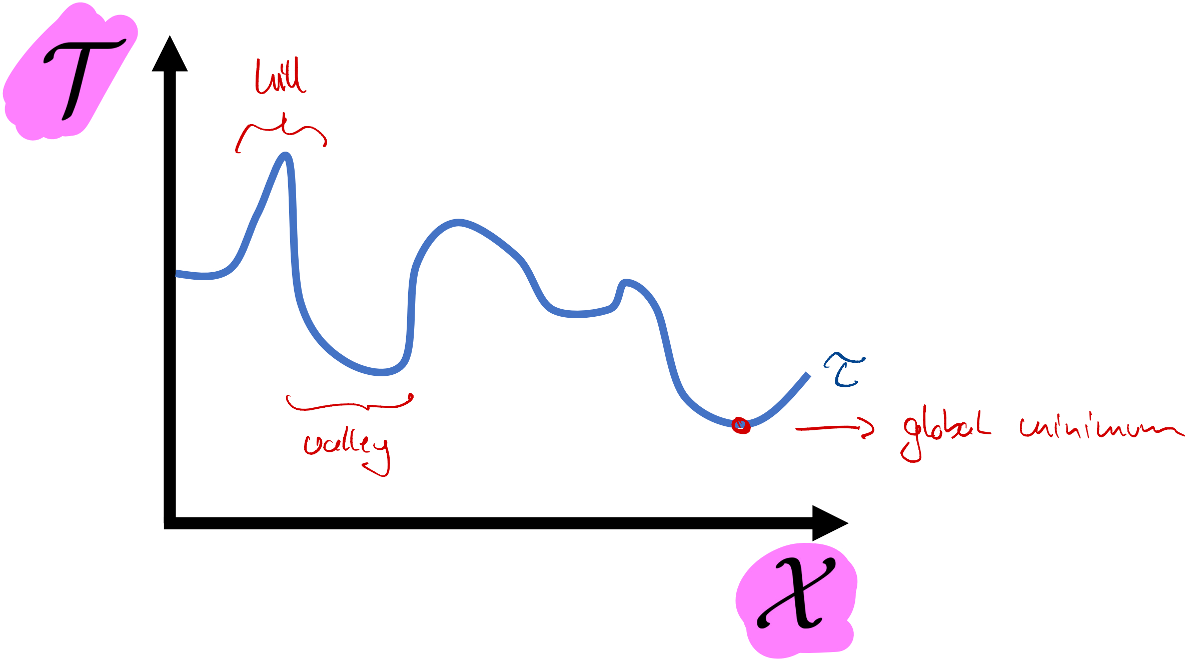
$\mathcal{J} = \mathbb{R}$ win ratio against baseline opponents

$\mathcal{J}(x) =$ use weights from x to battle n opponents;
return win rate

Further Examples

- training of neural networks (for AI)
- scheduling or logistics
- airport gate assignment
- portfolio optimization for finance
- ...

The Solution Landscape Metaphor



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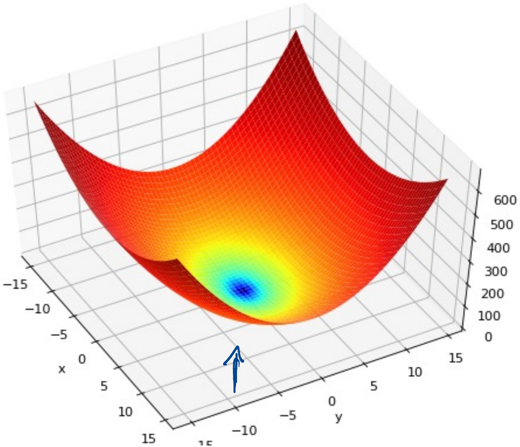
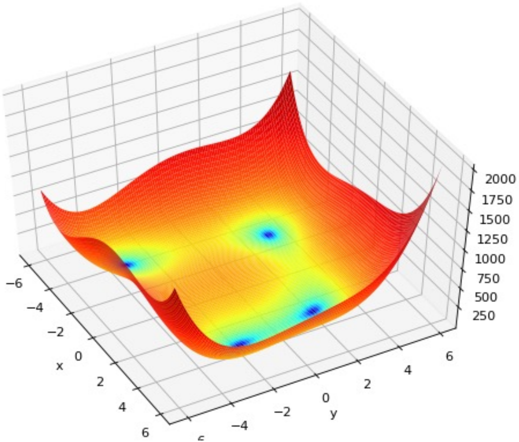


$\mathcal{J}$

$\mathcal{X} = \mathbb{R}^2$



source: deap.readthedocs.io/en/master/api/benchmarks.html

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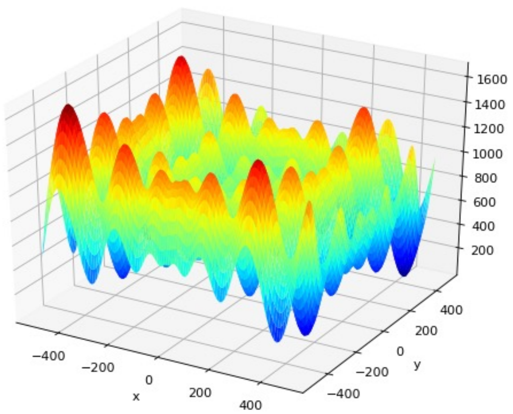

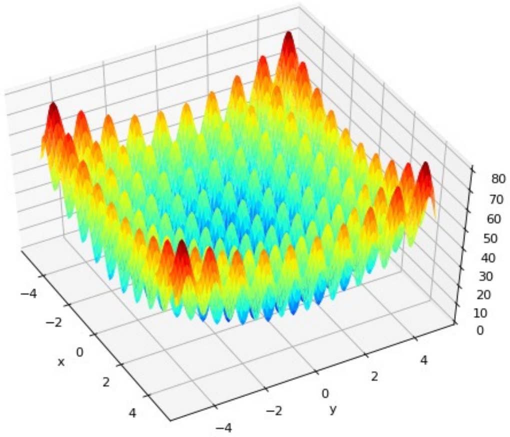

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Definition 9 (optimization). Let $\mathcal{X}$ be an arbitrary state space. Let $\mathcal{T}$ be an arbitrary set called target space and let $\leq$ be a total order on $\mathcal{T}$. A total function $\tau : \mathcal{X} \to \mathcal{T}$ is called target function. Optimization (minimization/maximization) is the procedure of searching for an $x \in \mathcal{X}$ so that $\tau(x)$ is optimal (minimal/maximal). Unless stated otherwise, we assume minimization. An optimization run of length $g+1$ is a sequence of states $\langle x_t \rangle_{0 \leq t \leq g}$ with $x_t \in \mathcal{X}$ for all t.

prev. lecture

Definition 9 (optimization). Let $\mathcal{X}$ be an arbitrary state space. Let $\mathcal{T}$ be an arbitrary set called target space and let $\leq$ be a total order on $\mathcal{T}$. A total function $\tau : \mathcal{X} \to \mathcal{T}$ is called target function. Optimization (minimization/maximization) is the procedure of searching for an $x \in \mathcal{X}$ so that $\tau(x)$ is optimal (minimal/maximal). Unless stated otherwise, we assume minimization. An optimization run of length $g+1$ is a sequence of states $\langle x_t \rangle_{0 \leq t \leq g}$ with $x_t \in \mathcal{X}$ for all t.

We use the following terminology:

- An optimization run is *working to some extent* if $t \leq t'$ for $t, t' \in [1; g] \subset \mathbb{N}$ correlates positively to some extent with $\tau(x_t) \geq \tau(x_{t'})$.
- An optimization run is *improving* if $\tau(x_t) \geq \tau(x_{t'})$ for all $t \leq t'$.
- An optimization run has *found a global optimum* iff $\tau(x) \geq \tau(x_g)$ for all $x \in \mathcal{X}$.

Let $e : \langle \mathcal{X} \rangle \times (\mathcal{X} \to \mathcal{T}) \to \mathcal{X}$ be a possibly randomized or non-deterministic function so that the optimization run $\langle x_t \rangle_{0 \le t \le g}$ is produced by calling e repeatedly, i.e., $x_{t+1} = e(\langle x_u \rangle_{0 \le u \le t}, \tau)$ for all t, $1 \le t \le g$, where x_0 is given externally (e.g., $x_0 =_{def} 42$) or chosen randomly (e.g., $x_0 \sim \mathcal{X}$). An optimization process is a tuple $(\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \le t \le g})$.

We use the following terminology:

- The *end result* of an optimization process $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \le t \le g})$ is x_g.
- The *best result* of an optimization process $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \le t \le g})$ is

$$ {}^*\mathcal{D} = \underset{\substack{x_t \\ 0 \le t \le g}}{\arg\min}\, \tau(x_t) $$

  with the *best target value* $|\mathcal{D}| = \tau({}^*\mathcal{D})$.
- An optimization process $\mathcal{D}' = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x'_t \rangle_{0 \le t \le g'})$ is a *continuation* of an optimization process $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \le t \le g})$ iff $g' \ge g$ and $x'_t = x_t$ for all t with $0 \le t \le g$. We write $\ell(\mathcal{D})$ for the set of all possible continuations of $\mathcal{D}$, which can be large if e is randomized or non-deterministic.
- An optimization process $\mathcal{D}$ is *elitist* iff for all its continuations $\mathcal{D}' = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x'_t \rangle_{0 \le t \le g'}) \in \ell(\mathcal{D})$ it holds that $\tau(x'_t) \ge \tau(x'_{t'})$ for all $t \le t'$.

How do we optimize for τ?

Algorithm 1 (single-sample random search)**.** Let $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_u \rangle_{0 \leq u \leq t})$ be an optimization process. The process $\mathcal{D}$ continues via (single-sample) random search if e is of the form

$$e(\langle x_u \rangle_{0 \leq u \leq t}, \tau) = x_{t+1} = \underset{x \in \{x_t, x'_t\}}{\arg\min}\ \tau(x)$$

where $x'_t \sim \mathcal{X}$ is drawn at random for each call to e.

Algorithm 2 (stochastic hill climbing)**.** Let $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_u \rangle_{0 \leq u \leq t})$ be an optimization process. Let $neighbors : \mathcal{X} \to \wp(\mathcal{X})$ be a function that returns a set of neighbors of a given state $x \in \mathcal{X}$. The process $\mathcal{D}$ continues via stochastic hill climbing if e is of the form

$$e(\langle x_u \rangle_{0 \leq u \leq t}, \tau) = x_{t+1} = \underset{x \in \{x_t, x'_t\}}{\arg\min}\ \tau(x)$$

where $x'_t \sim neighbors(x_t)$ is drawn at random for each call to e.

Algorithm 3 (simulated annealing)**.** Let $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_u \rangle_{0 \le u \le t})$ be an optimization process. Let $\mathit{neighbors} : \mathcal{X} \to \wp(\mathcal{X})$ be a function that returns a set of neighbors of a given state $x \in \mathcal{X}$. Let $\kappa : \mathbb{N} \to \mathbb{R}$ be a temperature schedule, i.e., a function that returns a temperature value for each time step. Let $A : \mathcal{T} \times \mathcal{T} \times \mathbb{R} \to \mathbb{P}$ with $\mathbb{P} = [0; 1] \subset \mathbb{R}$ be a function that returns an acceptance probability given two target values and a temperature. Commonly, we use

$$A(Q, Q', K) = \mathrm{e}^{\frac{-(Q'-Q)}{K}}$$

for $\mathcal{T} \subseteq \mathbb{R}$ and Euler's number e. The process $\mathcal{D}$ continues via simulated annealing if e is of the form

$$e(\langle x_u \rangle_{0 \le u \le t}, \tau) = x_{t+1} = \begin{cases} x'_t & \text{if } \tau(x'_t) \le \tau(x_t) \text{ or } r \le A(\underbrace{\tau(x_t)}_{Q}, \underbrace{\tau(x'_t)}_{Q'}, \underbrace{\kappa(t)}_{K}), \\ x_t & \text{otherwise,} \end{cases}$$

where $x'_t \sim \mathit{neighbors}(x_t)$ and $r \sim \mathbb{P}$ are drawn at random for each call to e.